



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL SYLLABUS

01-04-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	D	A	A	D	A	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	D	D	C	B	B	A	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	20	8	7	15	2	9	3	4	2	9

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	C	C	D	C	C	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	B	B	A	A	C	A	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	5	3	3	0	6	5	294	4	9

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	C	A	B	A	D	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	A	A	C	D	A	B	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	129	240	99	2	1	34	2	0	9	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A)

For projectile $\vec{p} \cdot \vec{f} = \vec{p} \cdot \vec{i}$

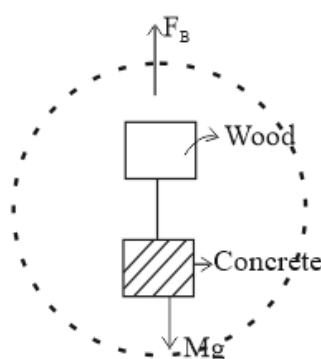
$$mV_3\hat{i} + mV_2\hat{j} - 2mV_1\hat{j} = 4m\vec{V}$$

$$mV_3\hat{i} + 2mV_1\hat{j} - 2mV_1\hat{j} = 4m\vec{V}$$

$$\Rightarrow \vec{V} = V_3/4\hat{i}$$

$$\therefore \text{Speed} = V_3/4$$

2. Ans (C)



$$F_B = \delta_w (V_1 + V_2) g$$

$$V_1 \rightarrow \text{Volume of wood}$$

$$V_2 \rightarrow \text{Volume of concrete}$$

$$m = m_1 + m_2$$

$$m_1 = \delta_{\text{wood}} V_1 \quad m_2 = \delta_{\text{concrete}} V_2$$

$$mg = F_B$$

$$\delta_{\text{wood}} V_1 g + \delta_{\text{concrete}} V_2 g = \delta_w (V_1 + V_2) g$$

$$\Rightarrow 0.5 \delta_w V_1 g + 2.5 \delta_w V_2 g = \delta_w (V_1 + V_2) g$$

$$\Rightarrow 0.5 V_1 + 2.5 V_2 = V_1 + V_2$$

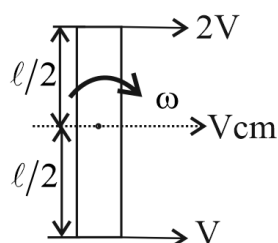
$$\Rightarrow 1.5 V_2 = 0.5 V_1$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{1.5}{0.5}$$

$$\frac{m_1}{m_2} = \frac{\delta_{\text{wood}} V_1}{\delta_{\text{concrete}} V_2} = \frac{0.5 \delta_w V_1}{2.5 \delta_w V_2}$$

$$= \frac{0.5}{2.5} \times \frac{1.5}{0.5} = \frac{3}{5}$$

3. Ans (A)



$$V_{\text{cm}} + \frac{w\ell}{2} = 2V \quad (1)$$

$$V_{\text{cm}} - \frac{w\ell}{2} = V \quad (2)$$

Adding (1) & (2)

$$2V_{\text{cm}} = 3V$$

$$V_{\text{cm}} = \frac{3V}{2}$$

$$W = (2V - V_{\text{cm}}) \times \frac{2}{\ell} = \left(2V - \frac{3V}{2}\right) \frac{2}{\ell}$$

$$= \frac{V}{2} \cdot \frac{2}{\ell} = \frac{V}{\ell}$$

$$\text{K.E} = \frac{1}{2} m v_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2$$

$$= \frac{1}{2} m \left(\frac{3V}{2}\right)^2 + \frac{1}{2} \cdot \frac{m\ell^2}{12} \cdot \frac{V^2}{\ell^2}$$

$$= \frac{9mV^2}{8} + \frac{mV^2}{24} = \frac{28mV^2}{24} = \frac{7}{6} mV^2$$

4. Ans (D)

$$P_i = P_{\text{atm}} + \frac{m_{\text{piston}} g}{A}$$

$$= 10^5 + \frac{1 \times 10}{1 \times 10^{-4}} = 2 \times 10^5 \text{ N/m}^2$$

$$P_f = P_{\text{atm}} + \frac{(m_{\text{piston}} + M_{\text{block}}) g}{A}$$

$$= 10^5 + \frac{63 \times 10}{1 \times 10^{-4}} = 64 \times 10^5 \text{ N/m}^2$$

For adiabatic process:

$$P_i^{1-\gamma} T_i^\gamma = P_f^{1-\gamma} T_f^\gamma$$

$$\gamma = \frac{5}{3}$$

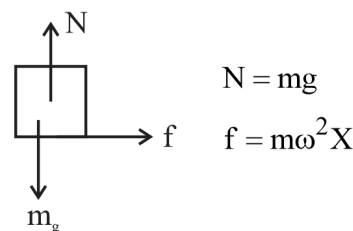
$$(2 \times 10^5)^{(1-5/3)} (400)^{5/3} = (64 \times 10^5)^{(1-5/3)} T_f^{5/3}$$

$$\frac{(400)^{5/3}}{(2 \times 10^5)^{2/3}} = \frac{(T_f)^{5/3}}{(64 \times 10^5)^{2/3}}$$

$$\Rightarrow T_f = (400) \left[\frac{64 \times 10^5}{2 \times 10^5} \right]^{2/5}$$

$$= 400 \times (2^5)^{2/5} = 400 \times 4 = 1600 \text{ K}$$

5. Ans (A)



$$N = mg$$

$$f = m\omega^2 X$$

$$\text{Maximum restoring force} = m\omega^2 A = f_{\text{max}} = \mu N$$

$$m\omega^2 A = \mu mg$$

$$\omega = \sqrt{\frac{\mu g}{A}}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{\mu g}{A}}$$

6. Ans (A)

$$1 \times 9.81 \text{ kg} \frac{\text{m}}{\text{s}^2} = x \text{ slug} \frac{(\text{ft})}{\text{s}^2}$$

$$9.81 \text{ N} = x \text{ pound}$$

$$= x (4.448) \text{ Newton}$$

$$\Rightarrow x = 2.205 \text{ pound}$$

7. Ans (D)

$$\frac{dV}{ds} = -\tan 60^\circ = -\sqrt{3}$$

$$a = V \frac{dV}{ds} = -4\sqrt{3} \text{ m/s}^2$$

$$V = 4 \text{ m/s.}$$

8. **Ans (A)**

$$\text{Initially, } \eta = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$$

$$\text{Finally, } \eta' = 1 - \frac{T_2 + \Delta T}{T_1 + \Delta T} = \frac{T_1 - T_2}{T_1 + \Delta T}$$

$$\eta' < \eta$$

9. **Ans (B)**

Component waves are :-

$$y_1 = \frac{A}{2} \sin(\omega t - kx) \text{ and } y_2 = \frac{A}{2} \sin(\omega t + kx)$$

$$y_1 + y_2 = A \sin \omega t \cos kx$$

$$B = \omega \text{ \& } C = k$$

Now, changed equation becomes :-

$$y_2' = \frac{A}{2} \sin(\omega t + 2\pi D t + k' x)$$

$$k' = \frac{\omega'}{V} = \frac{\omega + 2\pi D}{V} \text{ \& } k = \frac{\omega}{V}$$

$$k' = \frac{(\omega + 2\pi D) k}{\omega} = k + \frac{2\pi D C}{\omega}$$

$$= C + \frac{2\pi D C}{B}$$

$$y_2' = \frac{A}{2} \sin \left((B + 2\pi D) t + \left(C + \frac{2\pi D C}{B} \right) x \right)$$

10. **Ans (B)**

$$L.C = \frac{1 \text{ mm}}{50} = 0.02 \text{ mm}$$

$$\text{Zero error} = -(50 - 43) \times L.C$$

$$= -7 \times 0.02 = -0.14 \text{ mm}$$

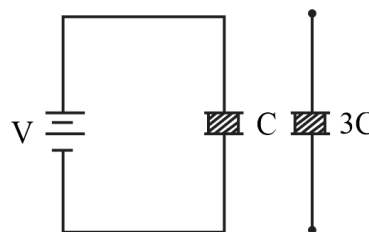
$$\text{Reading} = 4 \times 1 \text{ mm} + 0.02 \times 16 \text{ mm} + 0.14 \text{ mm}$$

$$= 4.46 \text{ mm.}$$

11. **Ans (D)**

$$U_i = \frac{1}{2} C V^2 + \frac{1}{2} (3C) V^2 = 2 C V^2$$

After introduction of dielectric:-



$$U_F = \frac{1}{2} (K C) V^2 + \frac{(3 C V)^2}{2 K (3 C)}$$

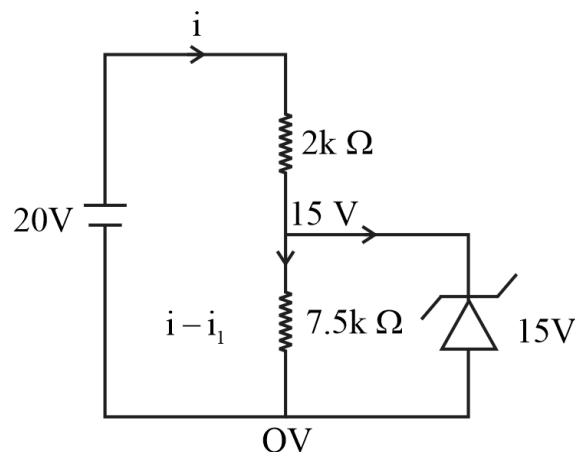
$$= \frac{1}{2} C V^2 \left(K + \frac{9}{3K} \right)$$

$$= \frac{1}{2} C V^2 \left(K + \frac{3}{K} \right)$$

$$\frac{U_f}{U_i} = \frac{\frac{1}{2} C V^2 \left(K + \frac{3}{K} \right)}{2 C V^2}$$

$$\frac{U_f}{U_i} = \frac{1}{4} \left(K + \frac{3}{K} \right)$$

12. **Ans (C)**



$$5 = 2 \times 10^3 i$$

$$\Rightarrow i = \frac{5}{2} \text{ mA} = 2.5 \text{ mA}$$

$$i - i_1 = \frac{15}{7.5 \times 10^3} = 2 \text{ mA}$$

$$i_1 = 0.5 \text{ mA}$$

13. Ans (D)

Fraction of total power carried by side bands

$$= \frac{m^2}{2 + m^2}$$

given $\frac{m^2}{2 + m^2} = \frac{2}{11}$

$$\Rightarrow 11 m^2 = 4 + 2 m^2$$

$$\Rightarrow 9 m^2 = 4$$

$$\Rightarrow m^2 = \frac{4}{9}$$

$$\Rightarrow m = \frac{2}{3}$$

$$\frac{A_m}{A_c} = m = \frac{2}{3}$$

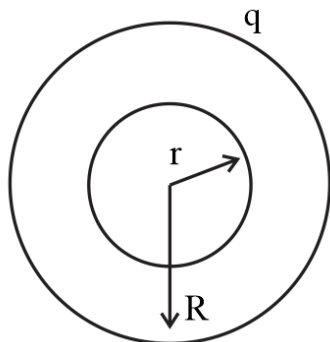
$$\Rightarrow A_m = \frac{2}{3} A_c$$

$$\Rightarrow A_m = \frac{2}{3} \times 66$$

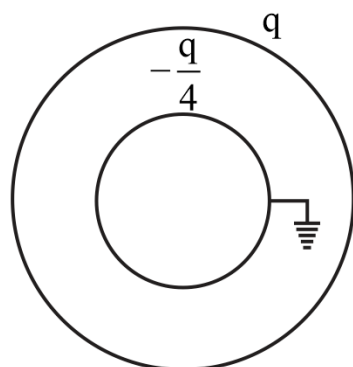
$$\Rightarrow A_m = 44$$

14. Ans (D)

Initially :-



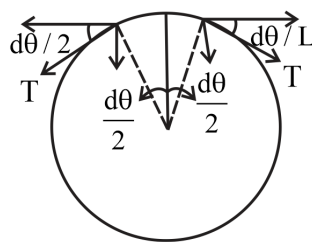
Finally :-



$$V_{\text{inner}} = \frac{Kq}{R} + \frac{K\left(\frac{-q}{4}\right)}{r} = 0$$

$$\Rightarrow R = 4r$$

15. Ans (C)



$$2T \sin\left(\frac{d\theta}{2}\right) = i(d\ell) B$$

$$\frac{d\theta}{2} \rightarrow 0 \therefore \sin \frac{d\theta}{2} = \frac{d\theta}{2}$$

$$\therefore 2T \frac{d\theta}{2} = iR(d\theta) B$$

$$\Rightarrow T = iRB$$

$$\Rightarrow 1.57 \times \left(\frac{0.5}{2\pi}\right) \times 1$$

$$= \frac{1.57 \times 0.5 \times 1}{2 \times 3.14} = \frac{1}{8} \text{ N} = 0.125 \text{ N}$$

16. Ans (B)

$$I = I_0 (1 - e^{-Rt/L})$$

$$0.5 I_0 = I_0 (1 - e^{-Rt/L})$$

$$e^{-Rt/L} = \frac{1}{2}$$

$$e^{Rt/L} = 2$$

$$\frac{Rt}{L} = \ln 2 = 0.7$$

$$\frac{R}{L} \times 1.11 = 0.7$$

$$\Rightarrow \frac{L}{R} = \frac{1.11}{0.7} = \frac{111}{70} \quad \text{----- (1)}$$

Energy stored in steady state

$$u = \frac{1}{2} L I_0^2$$

$$640 = \frac{1}{2} L \left(\frac{V}{R}\right)^2$$

$$640 = \frac{1}{2} \times \frac{L}{R^2} \times (200)^2$$

$$\Rightarrow \frac{L}{R^2} = \frac{640 \times 2}{4 \times 100 \times 10} = \frac{32}{1000} \quad \text{----- (2)}$$

(1)/(2)

$$R = \frac{111}{70 \times 32} \times 1000 = 49.56 \Omega$$

From (1) $\frac{L}{R} = \frac{111}{70}$

$$L = \frac{111}{70} \times 49.56$$

$$= 78.6 \text{ H}$$

17. Ans (B)

$$I = \frac{1}{2} c \epsilon_0 E_0^2 = \frac{\text{Power}}{\text{Area}}$$

$$\frac{1}{2} \times 3 \times 10^8 \times 8.85 \times 10^{-12} \times E_0^2 = \frac{600}{10^{-9}}$$

$$\Rightarrow E_0^2 = \frac{2 \times 600 \times 10^9}{3 \times 8.85 \times 10^{-4}}$$

$$= \frac{400}{8.85} \times 10^{13}$$

$$\Rightarrow E_0 \approx 2.12 \times 10^7 \text{ V/m}$$

18. Ans (A)

For single slit diffraction, angular width of central maxima is

$$\theta = \frac{2\lambda}{b} \quad b \rightarrow \text{slit width}$$

In YDSE angular fringe width

$$\phi = \frac{\lambda}{d} \quad d \rightarrow \text{distance between slits.}$$

Given, $d = 6.1 b$

$$\frac{d}{b} = 6.1$$

No. of maximas observed in central maxima of single slit diffraction pattern is

$$n = \frac{\theta}{\phi} = \frac{2\lambda/b}{\lambda/d} = 2 \left(\frac{d}{b} \right) = 2 \times 6.1 \approx 12$$

19. Ans (B)

$$((K.E)_{\text{maximum}})_1 = e v_1 = h v_1 - \phi$$

$$\text{_____ (1)}$$

$$((K.E)_{\text{maximum}})_2 = e v_2 = h v_2 - \phi$$

$$\text{_____ (2)}$$

$$(2) - (1)$$

$$e(v_2 - v_1) = h(v_2 - v_1)$$

$$\Rightarrow h = \frac{e(v_2 - v_1)}{v_2 - v_1} = \frac{1.6 \times 10^{-19} \times 9.9}{2.4 \times 10^{15}}$$

$$= 6.6 \times 10^{-34} \text{ Js}$$

$$= \frac{6.6 \times 10^{-34}}{1.6 \times 10^{-19}} = 4.125 \times 10^{-15} \text{ eVs}$$

20. Ans (B)

$$49 = 13.6 z^2 \left(\frac{1}{1^2} - \frac{1}{3^2} \right)$$

$$\Rightarrow \frac{49}{13.6} = z^2 \times \frac{8}{9}$$

$$\Rightarrow z^2 = \frac{49 \times 9}{13.6 \times 8}$$

$$\Rightarrow z^2 \approx 4 \Rightarrow z = 2$$

$$r_n = \frac{r_B n^2}{z} = r_B \frac{3^2}{2} = 4.5 r_B$$

$$2\pi r_n = n\lambda = \lambda = \frac{2\pi}{3} \times \frac{9}{2} r_B$$

PART-1 : PHYSICS

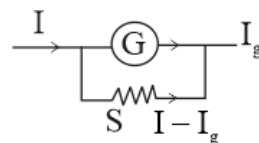
SECTION-II

1. Ans (20)

$$I = \frac{10}{10} = 1 \text{ A}$$

$$V = 10 - 1 \times 0.5 = 9.5 \text{ V}$$

For converting galvanometer to ammeter:-



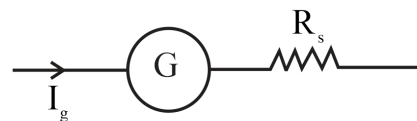
$$I_g R_G = (I - I_g) S$$

$$I_g R_G = IS - I_g S$$

$$I_g (R_G + 0.125) = 1 \times 0.125$$

$$\Rightarrow I_g = \frac{0.125}{(R_G + 0.125)} \text{ ---- (I)}$$

For converting galvanometer to voltmeter:-



$$I_g (R_G + R_s) = 9.5$$

$$I_g R_G + I_g 47 = 9.5$$

$$I_g (R_G + 47) = 9.5$$

$$\text{From (I)} \frac{0.125}{(R_G + 0.125)} (R_G + 47) = 9.5$$

$$\frac{8(R_G + 47)}{(8R_G + 1)} \times \frac{1}{8} = 9.5$$

$$R_G + 47 = 76 R_G + 9.5$$

$$75 R_G = 37.5 \Rightarrow R_G = 0.5 \Omega$$

$$I_g = \frac{0.125}{0.5 + 0.125} = \frac{0.125}{0.625} = 0.2 \text{ A}$$

$$\text{Power} = I_g^2 R_G = 0.04 \times 0.5$$

$$= 0.02 \text{ W} = 20 \text{ mW}$$

2. **Ans (8)**

$$\left(T_{1/2}\right)_x = \frac{1}{4} \text{Hr} \quad \left(T_{1/2}\right)_y = \frac{1}{6} \text{Hr}$$

$$\frac{1}{\left(T_{1/2}\right)_{xy}} = \frac{1}{\left(T_{1/2}\right)_x} + \frac{1}{\left(T_{1/2}\right)_y} = 10$$

$$\Rightarrow \left(T_{1/2}\right)_{xy} = \frac{1}{10} \text{Hr}$$

$$\text{After } \frac{1}{2} \text{Hr} \quad N = N_0 \left(\frac{1}{2}\right)^2$$

$$\text{After next 1 Hr} \quad N = N_0 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^6$$

$$\text{After another } \frac{1}{2} \text{Hr} \quad N = N_0 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^5$$

$$\Rightarrow N = \frac{N_0}{32} \left(\frac{1}{2}\right)^8 \Rightarrow n = 8.$$

3. **Ans (7)**

$$\frac{X_L - X_C}{R} = \tan 30^\circ \text{ or } -\tan 30^\circ$$

$$\Rightarrow \frac{X_L - X_C}{R} = \frac{1}{\sqrt{3}} \text{ or } \frac{-1}{\sqrt{3}} \text{ --- (1)}$$

Now, if L & C are doubled

$$\frac{2X_L - \frac{X_C}{2}}{R} = \sqrt{3} \text{ or } -\sqrt{3} \text{ --- (2)}$$

$$\frac{(1) \ 2(X_L - X_C)}{(2) \ 4X_L - X_C} = \frac{1}{3}$$

$$\Rightarrow 6X_L - 6X_C = 4X_L - X_C$$

$$\Rightarrow 2X_L = 5X_C$$

$$\text{or } \frac{2(X_L - X_C)}{4X_L - X_C} = \frac{-1}{3}$$

$$6X_L - 6X_C = 4X_L + X_C$$

$$10X_L = 7X_C$$

If L & C made x times

$$xX_L = \frac{X_C}{x}$$

$$\Rightarrow x^2 = \frac{X_C}{X_L} = \frac{2}{5} \text{ or } \frac{10}{7}$$

$$x = \sqrt{\frac{2}{5}} \text{ or } x = \sqrt{\frac{10}{7}}$$

$$\text{product } p = \sqrt{\frac{2}{5}} \sqrt{\frac{10}{7}}$$

$$p^2 = \frac{2}{5} \times \frac{10}{7} = \frac{4}{7}$$

$$\frac{1}{p^2} = \frac{7}{4} = 1.75$$

4. **Ans (15)**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v_1} + \frac{1}{-25} = \frac{1}{f} \text{ --- (1)}$$

$$\frac{1}{v_2} + \frac{1}{-20} = \frac{1}{f} \text{ --- (2)}$$

$$m_1 = \frac{-v_1}{-25} \quad m_2 = \frac{-v_2}{-20}$$

$$\frac{m_2}{m_1} = \frac{8}{7}$$

$$\frac{v_2}{20} \times \frac{25}{v_1} = \frac{8}{7}$$

$$\Rightarrow v_2 = \frac{32}{35} v_1$$

From (2)

$$\frac{35}{32v_1} - \frac{1}{20} = \frac{1}{f} \text{ --- (3)}$$

$$(1) - \frac{32}{35} (3)$$

$$-\frac{1}{25} + \frac{32}{35} \times \frac{1}{20} = \frac{1}{f} - \frac{32}{35f}$$

$$\Rightarrow \frac{-7+8}{7 \times 25} = \frac{3}{35f}$$

$$\Rightarrow \frac{1}{7 \times 25} = \frac{3}{35f}$$

$$\Rightarrow f = 15 \text{ cm.}$$

5. **Ans (2)**

$$i = \frac{8}{2+x}$$

$$5 - 2i = 1$$

$$\Rightarrow i = 2A$$

$$\Rightarrow 2 = \frac{8}{2+X}$$

$$\Rightarrow 2 + X = 4 \Rightarrow X = 2\Omega$$

6. **Ans (9)**

Let V_1 & V_2 be initial velocities of 1st & 2nd trolley

$$(140 + 40) V_1 = 60V_2$$

$$\Rightarrow 3V_1 = V_2$$

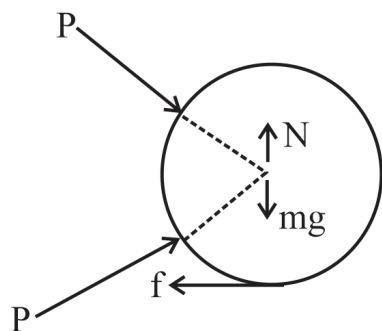
$$V_1^2 = 2aX_1 \text{ \& } V_2^2 = 2ax$$

$$\Rightarrow \frac{X_2}{X_1} = \frac{V_2^2}{V_1^2} = 9$$

$$\Rightarrow X_2 = 9X_1$$

$$\Rightarrow n = 9$$

7. Ans (3)



$$W_p + W_g + W_n + W_f = \Delta K.E$$

$$W_g = W_n = W_f = 0$$

$$\therefore W_p = 2p \cos 30^\circ (5.5) = \Delta K.E$$

$$5.5 \times 2 \times 425 \times \frac{\sqrt{3}}{2} = \frac{1}{2}mv^2 \left(1 + \frac{I}{mR^2}\right)$$

$$5.5 \times 425\sqrt{3} = \frac{1}{2} \times 600 \times V^2 \left(1 + \frac{1}{2}\right)$$

$$5.5 \times 425\sqrt{3} = \frac{3}{4} \times 600 V^2$$

$$V^2 = \frac{4 \times 425 \times 5.5}{600\sqrt{3}} \approx 9$$

$$V = 3\text{m/s}$$

8. Ans (4)

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{14 - 6}{14 + 6} = \frac{8}{20} = 0.4$$

9. Ans (2)

$$V = 4(1 - \cos 2x) \text{ J}$$

$$F = \frac{-dV}{dx} = -8 \sin 2x$$

$$a = F/m = -8 \sin 2x$$

$$\text{For small oscillations } \sin 2x \approx 2x$$

$$a = -16x = -\omega^2 x$$

$$\omega = 4$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ s}$$

10. Ans (9)

$$\text{Initially } V = 0, F_V = 0$$

$$F_{\text{mt}} = ma = mg = \rho_{\text{liq}} \cdot V_g$$

$$\rho v a = (\rho \cdot \rho_{\text{liq}}) V_g$$

$$a = \left(\frac{\rho - \rho_{\text{liq}}}{\rho} \right) g$$

$$a_{\text{avg}} = \frac{a}{2} = \frac{1}{2} \left(\frac{\rho - \rho_{\text{liq}}}{\rho} \right) g$$

At terminal velocity:

$$mg = F_B + F_V$$

$$\rho v g = \rho_{\text{liq}} \cdot v g + 6\pi\eta r v_T$$

$v_T \rightarrow$ Terminal velocity

$$V_T = \frac{(\rho - \rho_{\text{liq}}) \frac{2}{3} \pi r^3 g}{\eta} = \frac{2}{9} \frac{r^2 g}{\eta} (\rho - \rho_{\text{liq}})$$

$$a_{\text{avg}} = \frac{v_T - 0}{t} \quad t \rightarrow \text{time taken to reach}$$

terminal velocity

$$t = \frac{v_T}{a_{\text{avg}}} = \frac{(\rho - \rho_{\text{liq}}) \cdot \frac{2r^2 g}{9\eta}}{[(\rho - \rho_{\text{liq}})/\rho] \frac{g}{2}} = \frac{4}{9} \frac{r^2 \rho}{\eta}$$

$$\Rightarrow N = 9$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B)

$$\text{Volume strength} = 15 \times 10 = 150 \text{ ml.}$$

2. Ans (D)

$$E^\circ_{\text{cell 1}} = E^\circ_{\text{Cu}^+/\text{Cu}} - E^\circ_{\text{Cu}^{2+}/\text{Cu}^+} = 0.5$$

$$E^\circ_{\text{cell 2}} = E^\circ_{\text{Cu}^+/\text{Cu}} - E^\circ_{\text{Cu}^{2+}/\text{Cu}}$$

$$= E^\circ_{\text{Cu}^+/\text{Cu}} - \frac{E^\circ_{\text{Cu}^{2+}/\text{Cu}^+} + E^\circ_{\text{Cu}^+/\text{Cu}}}{2}$$

$$= \frac{E^\circ_{\text{Cu}^+/\text{Cu}} - E^\circ_{\text{Cu}^{2+}/\text{Cu}^+}}{2} = \frac{0.5}{2}$$

$$= 0.25 \text{ V}$$

3. Ans (C)

high T $\Rightarrow$ high entropy

4. **Ans (C)**

$$\frac{-1}{2} \frac{dB}{dt} = K [B]$$

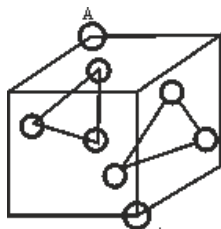
$$\int_{B_0}^{B_t} \frac{d[B]}{[B]} = -2k \int_0^t dt$$

$$2.303 \log \frac{[B_0]}{[B_t]} = 2kt$$

$$k = \frac{2.303}{40} \log 8 = \frac{2.303}{40} \times 0.9$$

$$= 0.052$$

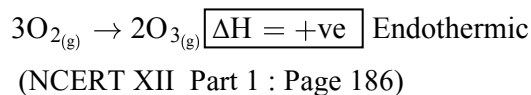
5. **Ans (C)**



$$d = \sqrt{3}a = \sqrt{3} (2\sqrt{2}r)$$

$$= 19.59 \text{ \AA}$$

7. **Ans (C)**



8. **Ans (C)**

3, (d), (q)

9. **Ans (D)**

A - Synthesis Gas

B - Producer Gas

10. **Ans (D)**

Li is the strongest reducing agent as it is due to its high hydration enthalpy.

11. **Ans (D)**

(d) (iv) (r)

12. **Ans (B)**

It is H_3BO_3 and it is a weak monobasic acid

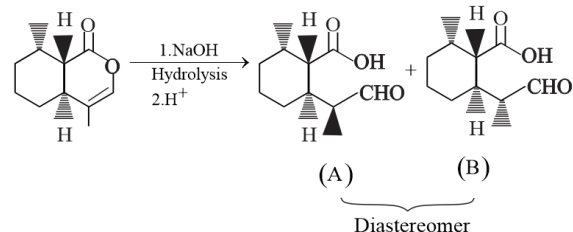
13. **Ans (B)**

A - NO_2

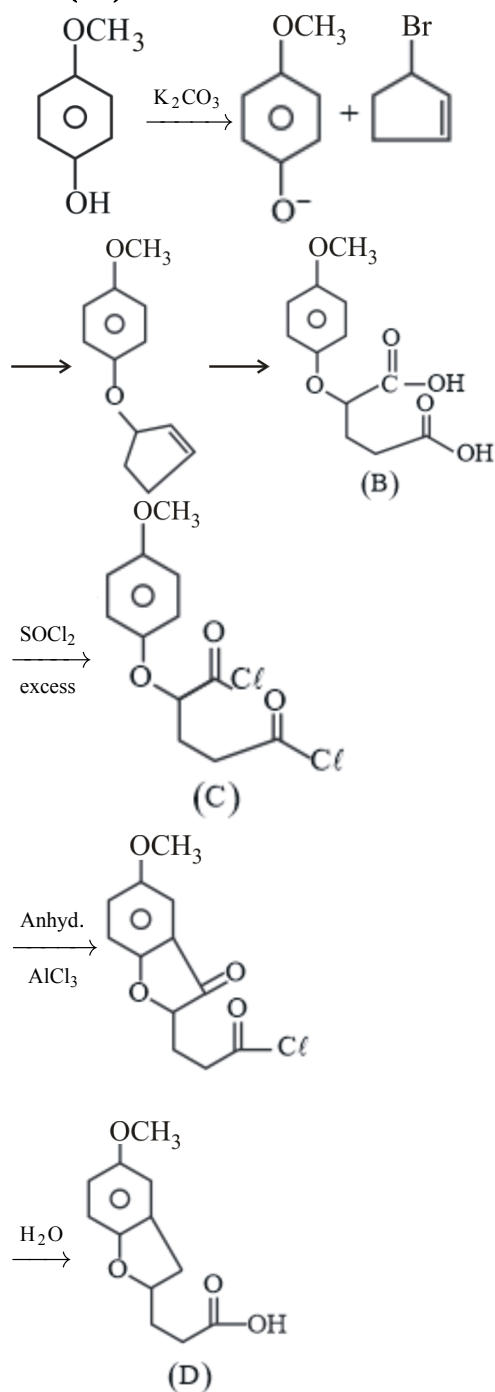
B - O_2

Both are paramagnetic

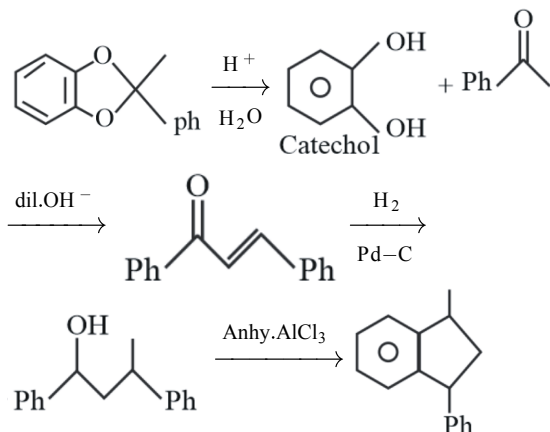
14. **Ans (B)**



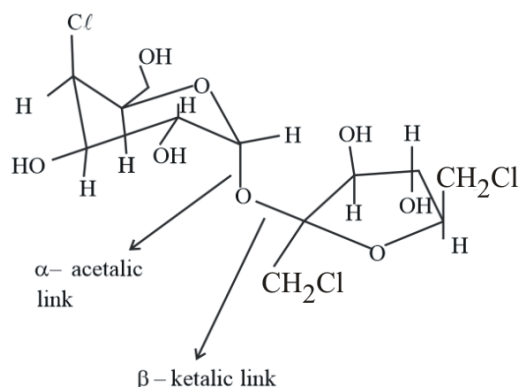
15. **Ans (A)**



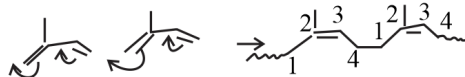
16. Ans (A)



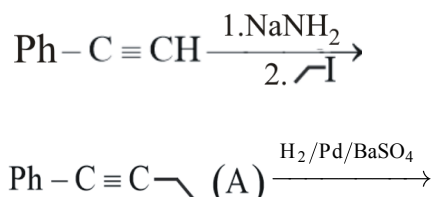
17. Ans (C)



18. Ans (A)



19. Ans (A)



20. Ans (B)

$$\begin{aligned}
 \Lambda_{\text{m}}^0(\text{HAc}) &= \lambda_{\text{H}^+}^0 + \lambda_{\text{Ac}^-}^0 \\
 &= \lambda_{\text{H}^+}^0 + \lambda_{\text{Cl}^-}^0 + \lambda_{\text{Ac}^-}^0 + \lambda_{\text{Na}^+}^0 - \lambda_{\text{Cl}^-}^0 - \lambda_{\text{Na}^+}^0 \\
 &= \Lambda_{\text{m}}^0(\text{HCl}) + \Lambda_{\text{m}}^0(\text{NaAc}) - \Lambda_{\text{m}}^0(\text{NaCl}) \\
 &= (425.9 + 91.0 - 126.4) \text{ S cm}^2 \text{ mol}^{-1} \\
 &= 390.5 \text{ S cm}^2 \text{ mol}^{-1}
 \end{aligned}$$

PART-2 : CHEMISTRY

SECTION-II

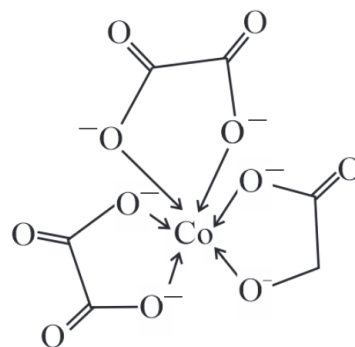
1. Ans (3)

$$x = 15$$

$$y = 2$$

$$z = 3$$

$$\frac{15}{3+2} = 3$$



2. Ans (5)

X is sp

Y is sp²

Z is sp²

$$\therefore x + y + z = 1 + 2 + 2 = 5.$$

3. Ans (3)

$$a = 3; b = 2; c = 3; d = 6; e = 1$$

$$\text{then } \frac{(6+3)2}{(3+2+1)} = 3$$

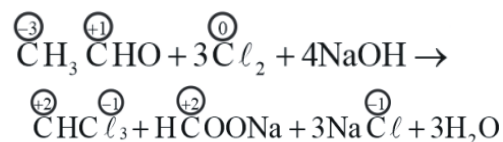
4. Ans (3)

(i), (ii), (iii) will produce NH₃ gas which gives white fumes other will produce N₂(g).

5. Ans (0)

W = 0 due to volume is constant.

6. Ans (6)



7. Ans (5)

$$E = n \cdot h\nu$$

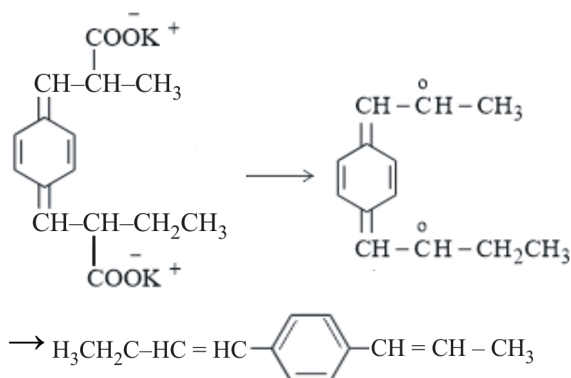
$$E_1 : E_2 = \bar{\nu}_1 : \bar{\nu}_2 = 2 : 3$$

8. Ans (294)

Molecular weight of aspartame

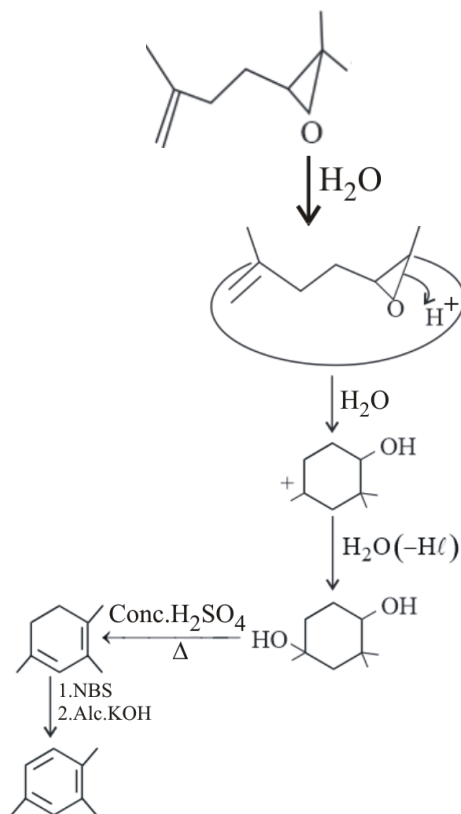
$$= \frac{100}{9.52} \times 14 \times 2 = 294.$$

9. Ans (4)



Total Cis - Trans isomers = 4

10. Ans (9)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

$$y = \frac{1}{x-2} = f(x)$$

$$xy - 2y = 1$$

$$x = \frac{2y+1}{y} = f^{-1}(y)$$

$$f^{-1}(x) = \frac{2x+1}{x}$$

For points of intersection, $f(x) = f^{-1}(x)$

$$\frac{1}{x-2} = \frac{2x+1}{x}$$

$$x = 2x^2 + x - 4x - 2$$

$$2x^2 - 4x - 2 = 0$$

$$x^2 - 2x - 1 = 0$$

$$x = 1 \pm \sqrt{2}$$

2. Ans (D)

Clearly domain is $[-1, 1]$

$\sin^{-1}x + \tan^{-1}x$ is increasing in $[-1, 1]$

and $\cos^{-1}x + \cot^{-1}x$ is decreasing in $[-1, 1]$

So, maximum value of the function occurs at

$$x = \pm 1$$

$$\left| \frac{\sin^{-1}1 + \tan^{-1}1}{\cos^{-1}1 + \cot^{-1}1} \right| = \left| \frac{\frac{\pi}{2} + \frac{\pi}{4}}{0 + \frac{\pi}{4}} \right| = 2 + 1 = 3$$

$$\left| \frac{\sin^{-1}(-1) + \tan^{-1}(-1)}{\cos^{-1}(-1) + \cot^{-1}(-1)} \right| = \left| \frac{-\frac{\pi}{2} - \frac{\pi}{4}}{\pi + \frac{3\pi}{4}} \right| = \frac{\frac{3\pi}{4}}{\frac{7\pi}{4}} = \frac{3}{7}$$

Clearly maximum value is 3.

3. Ans (B)

$$\lim_{x \rightarrow 0} \frac{\left(\frac{\sin^{-1}x}{x} \right)^n \cdot x^n \left(1 - 1 + \frac{x^2}{2} \right) \left(1 + x + \frac{x^2}{2} - 1 - x \right)}{\left(x - x + \frac{x^3}{3} \right) \left(x + \frac{x^3}{3} - x + \frac{x^3}{3} \right)}$$

$$\lim_{x \rightarrow 0} \frac{x^n \cdot \frac{x^2}{2} \cdot \frac{x^2}{2}}{\frac{x^3}{3} \cdot \frac{x^3}{3}} = \lim_{x \rightarrow 0} 3x^{n-2}$$

Clearly $n = 2$

4. **Ans (C)**

$$x^2y = c^3$$

Let us consider a point P $\left(ct, \frac{c}{t^2}\right)$ on the curve

$$2xy + x^2y' = 0; 2y + xy' = 0; y' = -\frac{2y}{x}$$

$$\text{Slope} = y' = -\frac{2}{ct} \cdot \frac{c}{t^2} = -\frac{2}{t^3}$$

$$\text{Eq. of line } y - \frac{c}{t^2} = -\frac{2}{t^3}(x - ct)$$

$$\text{Put } y = 0; \frac{ct}{2} = x - ct; x = \frac{3ct}{2} = a$$

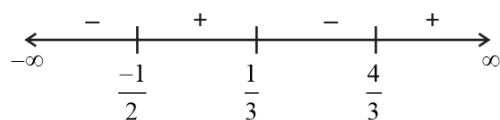
$$\text{Put } x = 0; y - \frac{c}{t^2} = \frac{2c}{t^2}; y = \frac{3c}{t^2} = b$$

$$a^2b = \frac{9c^2t^2}{4} \cdot \frac{3c}{t^2} = \frac{27c^3}{4}$$

5. **Ans (A)**

$$\text{Given, } f(x) = (2x + 1)^{50}(3x - 4)^{60}$$

$$f(x) = 220(2x + 1)^{49} \cdot (3x - 4)^{59} (3x - 1)$$



Sign scheme of $f(x)$

$\Rightarrow$ Least positive integer is $k = 2$.

6. **Ans (B)**

$$I = \int_{\frac{1}{2023}}^{2023} \frac{\tan^{-1} x}{x} dx \dots (1)$$

$$\text{Put, } x = \frac{1}{t}$$

$$I = \int_{2023}^{\frac{1}{2023}} \frac{\tan^{-1} \left(\frac{1}{t}\right)}{\left(\frac{1}{t}\right)} \left(-\frac{1}{t^2} dt\right)$$

$$I = \int_{\frac{1}{2023}}^{2023} \frac{\cot^{-1} x}{x} dx \dots (2)$$

Add (1) and (2)

$$2I = \int_{\frac{1}{2023}}^{2023} \frac{\pi/2}{x} dx = \frac{\pi}{2} (\log x)_{\frac{1}{2023}}^{2023}$$

$$I = \frac{\pi}{4} \left(\log 2023 - \log \frac{1}{2023} \right)$$

$$I = \frac{\pi}{4} 2 \log 2023 = \frac{\pi}{2} \log 2023.$$

7. **Ans (A)**

$$x = t^6$$

$$I = \int \frac{t^{-3} 6t^5 dt}{1 + t^2} = 6 \int \frac{t^2 dt}{1 + t^2}$$

$$= 6 \int \frac{1 + t^2 - 1}{1 + t^2} dt = 6 \int \left(1 - \frac{1}{1 + t^2} \right) dt$$

$$= 6 (t - \tan^{-1} t) + c$$

$$= 6 \left(x^{\frac{1}{6}} - \tan^{-1} \left(x^{\frac{1}{6}} \right) \right) + c$$

8. **Ans (D)**

$$I = \int_0^{\pi/4} \sin x (2\cos^2 x - 1) dx$$

Put $\cos x = t$

$$I = \int_1^{1/\sqrt{2}} (2t^2 - 1) (-dt) = \int_1^{1/\sqrt{2}} (1 - 2t^2) dt$$

$$I = \left(t - \frac{2t^3}{3} \right)_1^{1/\sqrt{2}} = \left(\frac{1}{\sqrt{2}} - \frac{2}{3} \cdot \frac{1}{2\sqrt{2}} \right) - \left(1 - \frac{2}{3} \right)$$

$$I = \frac{2}{3\sqrt{2}} - \frac{1}{3} = \frac{\sqrt{2} - 1}{3}.$$

$$I_1 = \int_0^{\pi/3} \sqrt{\sec x} \tan x dx = \int_0^{\pi/3} \frac{\sec x \tan x dx}{\sqrt{\sec x}}$$

Put $\sec x = t$

$$I_1 = \int_1^2 \frac{dt}{\sqrt{t}} = (2\sqrt{t})_1^2 = 2(\sqrt{2} - 1).$$

$$2(\sqrt{2} - 1) = k \frac{\sqrt{2} - 1}{3}$$

$$\boxed{k = 6}$$

9. Ans (B)

$$I_1 = \int_0^{\pi^2/4} \sin \sqrt{x} dx$$

Put $x = t^2$

$$I = \int_0^{\pi/2} \sin t \cdot 2t dt = 2(-t \cos t + \sin t) \Big|_0^{\pi/2}$$

$$\boxed{I = 2}$$

$$I_2 = \int_0^{\infty} \frac{dx}{(x + \sqrt{1+x^2})^2}$$

Put $x + \sqrt{1+x^2} = t$

$$1 + x^2 = (t - x)^2 = t^2 - 2xt + x^2$$

$$2xt = t^2 - 1$$

$$x = \frac{t^2 - 1}{2t} = \frac{1}{2} \left(t - \frac{1}{t} \right)$$

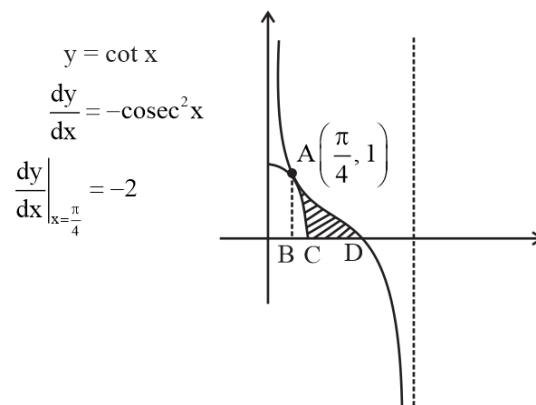
$$dx = \frac{1}{2} \left(1 + \frac{1}{t^2} \right) dt$$

$$I_2 = \int_1^{\infty} \frac{\frac{1}{2} \left(1 + \frac{1}{t^2} \right) dt}{t^2} = \frac{1}{2} \int_1^{\infty} \left(\frac{1}{t^2} + \frac{1}{t^4} \right) dt$$

$$I_2 = \frac{1}{2} \left(\frac{-1}{t} - \frac{1}{3t^3} \right) \Big|_1^{\infty} = \frac{1}{2} \left(0 + 1 + \frac{1}{3} \right) = \frac{2}{3}$$

$$\frac{I_1}{I_2} = \frac{2}{2/3} = 3.$$

10. Ans (B)



Equation of tangent at A $\left(\frac{\pi}{4}, 1 \right)$

$$y - 1 = -2 \left(x - \frac{\pi}{4} \right)$$

$$y = -2x + \frac{\pi}{2} + 1$$

Put $y = 0$, $x = \frac{\pi}{4} + \frac{1}{2}$ so C $\left(\frac{\pi}{4} + \frac{1}{2}, 0 \right)$

Area of region ACD = Area of region

ABD - Area of $\triangle ABC$

$$= \int_{\pi/4}^{\pi/2} \cot x - \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = \frac{1}{2} \ln 2 - \frac{1}{4}$$

11. **Ans (D)**

Solving both lines

$$3\lambda + 6 = 4\mu + 8$$

$$3\lambda - 4\mu = 2 \dots (i)$$

$$2\lambda + 4 = \mu + 2$$

$$2\lambda - \mu = -2 \dots (ii)$$

$$\Rightarrow \lambda = \mu = -2$$

P.O.I as lines L_1 and $L_2 \equiv (0, 0, 0)$.

Now, equation as plane

$$2x + y - z = 7$$

$$\Rightarrow 2(3\lambda + 6) + (2\lambda + 4) - (\lambda + 2) = 7$$

$$\Rightarrow \lambda = -1 \Rightarrow A \equiv (3, 2, 1)$$

Then for point B,

$$2(4\mu + 8) + (\mu + 2) - (2\mu + 4) = 7$$

$$\Rightarrow \mu = -1 \Rightarrow B \equiv (4, 1, 2)$$

Now, image as $(0, 0, 0)$ w.r.t plane

$$= \left(\frac{14}{3}, \frac{7}{3}, \frac{-7}{3} \right)$$

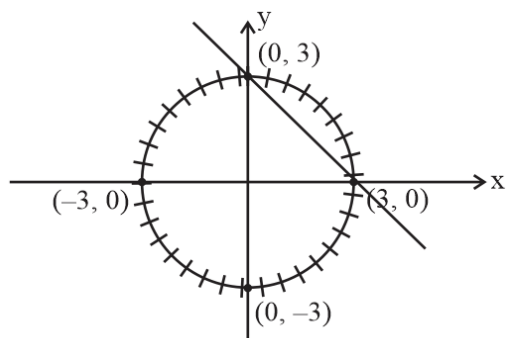
$\Rightarrow$ image of $(0, 0, 0)$ w.r.t AB is

$$\left(\frac{14}{3}, \frac{16}{3}, \frac{2}{3} \right).$$

$\Rightarrow$ equation of line joining these two points

$$= \frac{3x - 14}{0} = \frac{3y - 7}{3} = \frac{3z + 7}{3}.$$

12. **Ans (B)**



Now, $x + y \leq 3$

$$= 1(0, 0) \rightarrow -ve$$

$$\text{total length} = 2\pi r = 6\pi$$

$$\text{Fav. length} = \frac{3}{4}(6\pi)$$

$$P = \frac{3}{4}$$

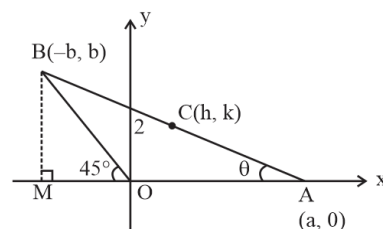
13. **Ans (A)**

$E_i \rightarrow i$ white and $10 - i$ black

$A \rightarrow 3$ black balls are drawn

$$\begin{aligned} P\left(\frac{E_1}{A}\right) &= \frac{P(E_1) \cdot P\left(\frac{A}{E_1}\right)}{\sum P(E_i) \cdot P\left(\frac{A}{E_i}\right)} \\ &= \frac{\frac{1}{11} \times \frac{{}^9C_3}{{}^{10}C_3}}{\frac{1}{11} \left(\frac{{}^{10}C_3 + {}^9C_3 + \dots + {}^3C_3}{{}^{10}C_3} \right)} \\ &= \frac{{}^9C_3 / {}^{10}C_3}{{}^{11}C_4 / {}^{10}C_3} \\ &= \frac{14}{55} \end{aligned}$$

14. **Ans (A)**



$$\Rightarrow BM = 2 \sin \theta, MA = 2 \cos \theta$$

$$\Rightarrow OM = 2 \sin \theta \Rightarrow B \equiv (-2 \sin \theta, 2 \sin \theta)$$

$$\Rightarrow A \equiv (2 \cos \theta - 2 \sin \theta, 0)$$

$$\text{Now, } h = \frac{2 \cos \theta - 2 \sin \theta - 2 \sin \theta}{2}$$

$$\Rightarrow \cos \theta - 2 \sin \theta = h \dots (i)$$

$$k = \frac{0 + 2 \sin \theta}{2} \Rightarrow \sin \theta = k \dots (ii)$$

From (i)

$$\Rightarrow \cos \theta = h + 2k \dots (iii)$$

$$(ii)^2 + (iii)^2$$

$$\Rightarrow 1 = k^2 + (h + 2k)^2$$

$$\Rightarrow x^2 + 5y^2 + 4xy = 1$$

15. **Ans (C)**

Let, centre $\equiv (h, k)$

$$\left| \frac{5h + 12k - 10}{13} \right| = 3 \text{ and } \left| \frac{5h - 12k - 40}{13} \right| = 3$$

$$\Rightarrow 5h + 12k - 10 = \pm 39 \dots (i)$$

$$\Rightarrow 5h - 12k - 40 = \pm 39 \dots (ii)$$

Solving (i) & (ii)

$$\Rightarrow h = 5, k = 2$$

16. Ans (D)

Both parabola are symm. about line $y = x$.
 $\Rightarrow$ Minimum distance is between tangents to parabola parallel to $y = x$.

$$\Rightarrow x^2 - 8x - 4y + 40 = 0$$

$\Rightarrow$ differentiate w.r.t x .

$$\Rightarrow 2x - 8 - 4y^1 = 0$$

$$\Rightarrow y^1 = \frac{x-4}{2} = 1 \Rightarrow x = 6$$

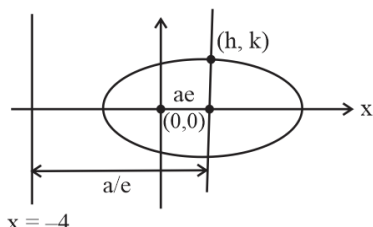
$$\Rightarrow y = 7 \Rightarrow P \equiv (6, 7)$$

Similarly, $Q \equiv (7, 6)$

$$d = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$d^2 = 2.$$

17. Ans (A)



$$\Rightarrow k = b, h = ae \text{ and } a/e - ae = 4$$

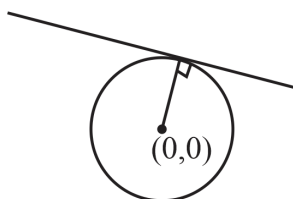
$$\Rightarrow a(1 - e^2) = 4e \Rightarrow a^2(1 - e^2) = 4ae$$

$$\Rightarrow b^2 = 4h$$

$$\Rightarrow k^2 = 4h \Rightarrow y^2 = 4x$$

18. Ans (B)

Using concept of homogenising,



$$\frac{x^2}{a^2} - \frac{y^2}{2a^2} - 1^2 = 0$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{2a^2} - \left(\frac{x \cos \alpha + y \sin \alpha}{P} \right)^2 = 0$$

$$\theta = 90^\circ$$

$$\Rightarrow \text{coefficient of } x^2 + \text{coefficient of } y^2 = 0$$

$$\Rightarrow \frac{1}{a^2} - \frac{\cos^2 \alpha}{P^2} - \frac{1}{2a^2} - \frac{\sin^2 \alpha}{P^2} = 0$$

$$\Rightarrow \frac{1}{2a^2} = \frac{1}{P^2} \Rightarrow P = \sqrt{2}a$$

$$\Rightarrow \text{radius} = \sqrt{2}a$$

19. Ans (C)

$$\text{Let } p \wedge q = m$$

$$mv \sim m = s$$

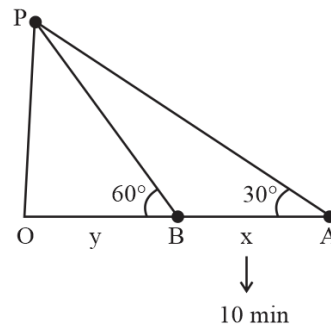
Hence option (C) is correct.

20. Ans (B)

$$\tan 60^\circ = \frac{h}{y} = \sqrt{3}$$

$$h = y\sqrt{3} \dots (i)$$

$$\tan 30^\circ = \frac{h}{x+y} = \frac{1}{\sqrt{3}}$$



$$x + y = h\sqrt{3} = (y\sqrt{3})3$$

$$\Rightarrow x + y = 3y \Rightarrow x = 2y$$

$$\Rightarrow y = \frac{x}{2} \Rightarrow \text{time taken} = \frac{10}{2} = 5 \text{ min.}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (129)

$$T_{r+1} = {}^{1024}C_r \left(5^{\frac{1}{2}}\right)^{1024-r} \left(7^{\frac{1}{8}}\right)^2$$

where $r = 0, 1, 2, \dots, 1024$.

Since T_{r+1} is an integer, r must be divisible by 8.

$$\Rightarrow r = 0, 8, 16, \dots, 1024$$

$$\Rightarrow 129 \text{ numbers.}$$

2. Ans (240)

By inclusion - exclusion principle,

$$\text{No. of ways} = 4^5 - {}^4C_1 \cdot 3^5 + {}^4C_2 \cdot 2^5 - {}^4C_1 \cdot 1^5$$

$$= 1024 - 972 + 192 - 4$$

$$= 240.$$

3. Ans (99)

Let, $x = 2^p 3^q$ where

$$p = 0, 1, \dots, 10 \Rightarrow 11 \text{ choices}$$

$$q = 0, 1, \dots, 8 \Rightarrow 9 \text{ choices}$$

Once x is selected, y gets automatically selected

$$\left(\because y = \frac{2^{10} 3^8}{x} \right).$$

$$\Rightarrow 11 \times 9 = 99 \text{ ways}$$

4. **Ans (2)**

$$\frac{\cos 37^\circ}{\sin 37^\circ} - \frac{\sin 37^\circ}{\cos 37^\circ} = \frac{\cos 74^\circ}{\sin 74^\circ} = \frac{\cos^2 37^\circ - \sin^2 37^\circ}{\sin 37^\circ \cos 37^\circ} = \frac{\cos^2 37^\circ - \sin^2 37^\circ}{2 \sin 37^\circ \cos 37^\circ} = 2$$

5. **Ans (1)**

$$\begin{aligned} & \sqrt{2 - \sqrt{2 - \sqrt{2 + 2 \cos 480^\circ}}} \\ &= \sqrt{2 - \sqrt{2 - \sqrt{4 \cos^2 240^\circ}}} \\ &= \sqrt{2 - \sqrt{2 + 2 \cos 240^\circ}} \\ &= \sqrt{2 - \sqrt{4 \cos^2 120^\circ}} \\ &= \sqrt{2 + 2 \cos 120^\circ} \\ &= \sqrt{4 \cos^2 60^\circ} \\ &= 1. \end{aligned}$$

6. **Ans (34)**

$$S = 1^2 + \frac{3^2}{2} + \frac{5^2}{2^2} + \frac{7^2}{2^3} + \dots$$

$$\frac{S}{2} = \frac{1^2}{2} + \frac{3^2}{2^2} + \frac{5^2}{2^3} + \dots$$

$$\frac{S}{2} = 1^2 + 2 \left(\frac{4}{2} + \frac{8}{2^2} + \frac{12}{2^3} + \dots \right)$$

$$\frac{S}{2} = 1^2 + 8 \left(\underbrace{\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots}_P \right) \dots (i)$$

$$P = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots$$

$$\frac{P}{2} = \frac{1}{2^2} + \frac{2}{2^3} + \dots$$

$$\frac{P}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{2}{2^3} + \dots = \frac{1/2}{1 - 1/2} = 1$$

$$\Rightarrow P = 2 \Rightarrow S = 34$$

7. **Ans (2)**

We have, for $n \geq 1$

$$\begin{aligned} t_n &= \sum_{r=1}^n t_r - \sum_{r=1}^{n-1} t_r \\ &= \frac{1}{6} n(n+1)(n+2 - n + 1) \\ &= \frac{1}{2} n(n+1) \end{aligned}$$

Thus,

$$\begin{aligned} \sum_{r=1}^n \frac{1}{t_r} &= 2 \sum_{r=1}^n \left[\frac{1}{r} - \frac{1}{r+1} \right] = 2 \left(1 - \frac{1}{n+1} \right) \\ \Rightarrow \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{t_r} &= \lim_{n \rightarrow \infty} 2 \left(1 - \frac{1}{n+1} \right) = 2(1 - 0) = 2. \end{aligned}$$

8. **Ans (0)**

$$x^2 = 3 - x, \quad x^3 = 3x - x^2 = 3x - (3 - x) = 4x - 3$$

$$\Rightarrow \alpha^3 - 4\beta^2 + 19 = 4\alpha - 3 - 4(3 - \beta) + 19$$

$$= 4(\alpha + \beta) + 4 = 0$$

9. **Ans (9)**

$$x, y, z \text{ are roots of } t^3 - 3t + p = 0$$

$$|x^3 y + y^3 z + z^3 x| = |(3x - p)y + (3y - p)z + (3z - p)x|$$

$$= |3(xy + yz + zx) - p(x + y + z)| = 9.$$

10. **Ans (5)**

$$a^3 + b^3 + c^3 = (a + b + c)^3$$

$$\Rightarrow 3(a + b)(b + c)(c + a) = 0$$

$$\Rightarrow \underbrace{\sin x + \cos x = 2}_{\text{No. solutions}} \text{ or } \underbrace{\sin x + \cos x = 1}_{x=0, \frac{\pi}{2}, 2\pi} \text{ or } \underbrace{2 \sin x = 1}_{x=\frac{\pi}{6}, \frac{5\pi}{6}}$$